

EOW Report – Coring

Company: Equinor
Field: Northern Lights
Rig: West Hercules
Well: 31/5-7 EOS

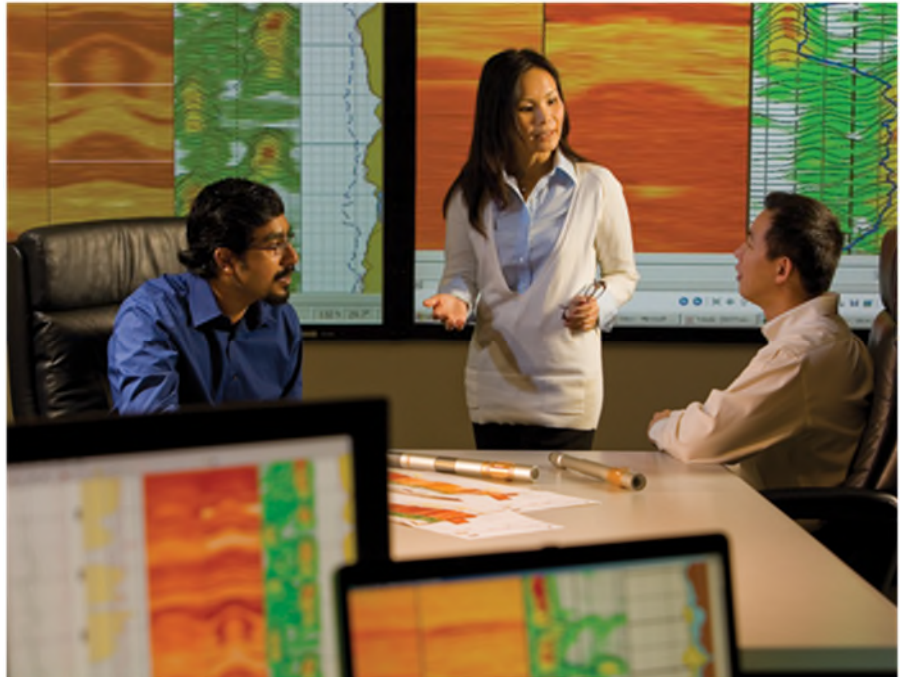


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1. CORING SERVICES SUMMARY

Customer:	Equinor	Area:	North Sea
Well Name:	31/5-7 EOS	Drilling Rig:	West Hercules
Country:	Norway	Date:	14.01.2020

Core No.	Top Depth	Bottom Depth	barrel Length	Cored	Recovered	Recovery
1	2592md	2596md	27m	4m	3.2m	80.0 %
2	2643md	2697md	54m	54m	53.4m	98.9 %
3	2709md	2746 md	81	37	35.3m	95.4 %
4	2747md	2784 md	36	36.7	33.9m	92.4 %

Comments:

A total of 131.7 meters of core was cut and 125.8 meters recovered, 95.5 % recovery, over 4 core runs on the recent EOS performed on West Hercules for Equinor.

The coring and core handling was performed by Baker Hughes coring services.

Additional Information:

Core barrel Information	
Outer barrel Type:	HT30 Max 7 1/4"– 5 15/16" Jambuster
Inner Tube Type:	Aluminium, vented
Pressure Vented:	Yes
Catcher Type:	Standard Spring Catcher
Core Head Run: 1 - 2 Core Head Run: 3 - 4	BHC 409Z (8 1/2" x 4") PDC BHC TC409 (8 1/2" x 4") PDC
Core OD:	4"

2. CORING BHA

BOTTOM HOLE ASSEMBLY Coring - 27M BRL											
									Date: December 31, 2019		
COMPANY Equinor				Well Name 31/5-7 EOS				Rig West Hercules		Field 0	
Core	Size	Mfr.	TYPE	Serial No.	Nozzles	TFA	Fish Neck	Bit L (m)			
1	7 1/4" x 5 15/16"	HC	BHC409Z	7160215	NA	1.50	0.21	0.43			
Item	Component	Down	Up	Vendor	Serial No.	Gauge OD	Body OD	Body ID	Fish Neck	Comp L (m)	
1	Corehead BHC409Z		HT-30	HC	7160215	8 1/2"	7 1/4"	4 "	0.21	0.45	0.45
2	Stabilizer	HT-30	HT-30	BHI	14378317	8 15/32"	7 1/4"	5 3/8"	0.51	1.22	1.67
3	Outertube	HT-30	HT-30	BHI	14348681		7 1/4"	5 3/8"	Slick	7.93	9.60
4	Stabilizer	HT-30	HT-30	BHI	14378315	8 15/32"	7 1/4"	5 3/8"	0.51	1.22	10.82
5	Outertube	HT-30	HT-30	BHI	14278066		7 1/4"	5 3/8"	Slick	7.93	18.75
6	Stabilizer	HT-30	HT-30	BHI	13970760	8 15/32"	7 1/4"	5 3/8"	0.51	1.22	19.97
7	Outertube	HT-30	HT-30	BHI	14477119		7 1/4"	5 3/8"	Slick	7.93	27.90
8	Stabilizer	HT-30	HT-30	BHI	14339941	8 15/32"	7 1/4"	5 3/8"	0.51	1.22	29.12
9	LDA ext.	HT-30	HT-30	BHI	14447792		7 1/4"	5 3/8"	Slick	1.22	30.34
10	Top sub W/float	HT-30	NC50	BHI	14426032		7 1/4"	3 5/32"	Slick	0.77	31.11
11	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39.42	70.53
12	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39.39	109.92
13	Jar	NC50	NC50	SLB	TALLY					12.10	122.02
14	3 x 6 1/2 collar	NC50	NC50	RIG	TALLY					27.99	150.01
15	XO	NC50	5 1/2FH	RIG	TALLY					1.00	151.01
16	12 X 5 7/8 HWDP	5 1/2FH	5 1/2FH	RIG	TALLY					110.57	261.58
17	5 7/8 DPS RANGE 3	5 1/2FH	5 1/2FH	RIG	TALLY					13.93	275.51
18											275.51
19											275.51
20											275.51
21											275.51
22											275.51
23											275.51

BOTTOM HOLE ASSEMBLY

Coring - 54M BRL

										Date: 01.01.2020	
COMPANY Equinor				Well Name 31/5-7 EOS				Rig West Hercules			Field Northern Lights
Core	Size	Mfr.	TYPE	Serial No.	Nozzles	TFA		Fish Neck		Bit L (m)	
2	7 1/4" x 5 15/16"	HC	BHC409Z	7160215	N/A	1.50		0.21		0.45	
Item	Component	Down	Up	Vendor	Serial No.	Gauge O	Body OD	Body ID	Fish Neck	Comp L (m)	
1	Corehead BHC409Z		HT-30	HC	7160215	8 1/2"	7 1/4"	4 "	0.21	0.45	0.45
2	Stabilizer	HT-30	HT-30	BHI	14378317	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	1.67
3	Outertube	HT-30	HT-30	BHI	14348681		7 1/4"	5 15/16"	Slick	7.93	9.60
4	Stabilizer	HT-30	HT-30	BHI	14378315	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	10.82
5	Outertube	HT-30	HT-30	BHI	14278066		7 1/4"	5 15/16"	Slick	7.93	18.75
6	Stabilizer	HT-30	HT-30	BHI	13970760	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	19.97
7	Outertube	HT-30	HT-30	BHI	13956501		7 1/4"	5 15/16"	Slick	7.93	27.90
8	Stabilizer	HT-30	HT-30	BHI	13982992	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	29.12
9	Outertube	HT-30	HT-30	BHI	14347351		7 1/4"	5 15/16"	Slick	7.93	37.05
10	Stabilizer	HT-30	HT-30	BHI	14409163	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	38.27
11	Outertube	HT-30	HT-30	BHI	13960478		7 1/4"	5 15/16"	Slick	7.93	46.20
12	Stabilizer	HT-30	HT-30	BHI	13982996	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	47.42
13	Outertube	HT-30	HT-30	BHI	14477119		7 1/4"	5 15/16"	Slick	7.93	55.35
14	Stabilizer	HT-30	HT-30	BHI	14339941	8 15/32"	7 1/4"	5 15/16"	0.51	1.22	56.57
15	LDA ext.	HT-30	HT-30	BHI	14447792		7 1/4"	5 15/16"	Slick	1.22	57.79
16	Top sub	HT-30	NC50	BHI	14426032		7 1/4"	3 5/32"	Slick	0.76	58.55
17	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39.42	97.97
18	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39.39	137.36
19	Jar	NC50	NC50	SLB	TALLY					12.10	149.46
20	3 x 6 1/2 collar	NC50	NC50	RIG	TALLY					27.99	177.45
21	XO	NC50	5 1/2FH	RIG	TALLY					1.00	178.45
22	12 X 5 7/8 HWD	5 1/2FH	5 1/2FH	RIG	TALLY					110.57	289.02
23	5 7/8 DPS RANGE 3	5 1/2FH	5 1/2FH	RIG	TALLY					13.93	302.95
24											302.95
25											302.95
26											302.95
27											302.95
28											302.95
29											302.95

BOTTOM HOLE ASSEMBLY Coring - 81M BRL

										Date: 10.01.2020	
COMPANY Equinor				Well Name 31/5-7 EOS				Rig West Hercules		Field Northern Lights	
Core	Size	Mfr.	TYPE	Serial No.	Nozzles	TFA	Fish Neck	Bit L (m)			
3	7 1/4" x 5 15/16"	HC	BHC409Z	5287192	N/A	1,50	0,21	0,45			
Item	Component	Down	Up	Vendor	Serial No.	Gauge OD	Body OD	Body ID	Fish Neck	Comp L (m)	
1	Corehead BHC409Z		HT-30	HC	5287192	8 1/2"	7 1/4"	4"	0,21	0,45	0,45
2	Stabilizer	HT-30	HT-30	BHI	14378317	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	1,67
3	Outertube	HT-30	HT-30	BHI	14348681		7 1/4"	5 15/16"	Slick	7,93	9,60
4	Stabilizer	HT-30	HT-30	BHI	14378315	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	10,82
5	Outertube	HT-30	HT-30	BHI	14278066		7 1/4"	5 15/16"	Slick	7,93	18,75
6	Stabilizer	HT-30	HT-30	BHI	13970760	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	19,97
7	Outertube	HT-30	HT-30	BHI	13956501		7 1/4"	5 15/16"	Slick	7,93	27,90
8	Stabilizer	HT-30	HT-30	BHI	13982992	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	29,12
9	Outertube	HT-30	HT-30	BHI	14347351		7 1/4"	5 15/16"	Slick	7,93	37,05
10	Stabilizer	HT-30	HT-30	BHI	14409163	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	38,27
11	Outertube	HT-30	HT-30	BHI	13960478		7 1/4"	5 15/16"	Slick	7,93	46,20
12	Stabilizer	HT-30	HT-30	BHI	13982996	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	47,42
13	Outertube	HT-30	HT-30	BHI	13933572		7 1/4"	5 15/16"	Slick	7,93	55,35
14	Stabilizer	HT-30	HT-30	BHI	17173336	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	56,57
15	Outertube	HT-30	HT-30	BHI	14348678		7 1/4"	5 15/16"	Slick	7,93	64,50
16	Stabilizer	HT-30	HT-30	BHI	14613091	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	65,72
17	Outertube	HT-30	HT-30	BHI	14348673		7 1/4"	5 15/16"	Slick	7,93	73,65
18	Stabilizer	HT-30	HT-30	BHI	13982995	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	74,87
19	Outertube	HT-30	HT-30	BHI	14477119		7 1/4"	5 15/16"	Slick	7,93	82,80
20	Stabilizer	HT-30	HT-30	BHI	14339941	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	84,02
21	LDA ext.	HT-30	HT-30	BHI	14447792		7 1/4"	5 15/16"	Slick	1,22	85,24
22	Top sub	HT-30	NC50	BHI	14428032		7 1/4"	3 5/8"	Slick	0,76	86,00
23	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39,42	125,42
24	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39,39	164,81
25	Jar	NC50	NC50	SLB	TALLY					12,10	176,91
26	3 x 6 1/2 collar	NC50	NC50	RIG	TALLY					27,99	204,90
27	XO	NC50	5 1/2FH	RIG	TALLY					1,00	205,90
28	12 X 5 7/8 HWDP	5 1/2FH	5 1/2FH	RIG	TALLY					110,57	316,47
29	5 7/8 DPS RANGE 3	5 1/2FH	5 1/2FH	RIG	TALLY					13,93	330,40
30											330,40
31											330,40
32											330,40
33											330,40
34											330,40
35											330,40

BOTTOM HOLE ASSEMBLY **Coring - 36 M BRL**

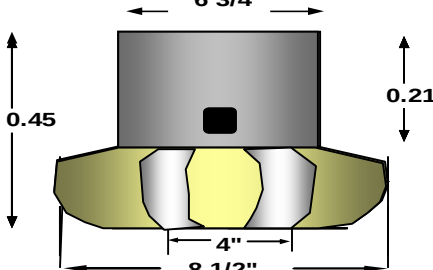
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COMPANY Equinor				Well Name 31/5-7 EOS			Rig West Hercules			Field Northern Lights	
Core	Size	Mfr.	TYPE	Serial No.	Nozzles	TFA	Fish Neck	Bit L (m)			
4	7 1/4" x 5 15/16"	HC	BHC409Z	5287192	N/A	1,50	0,21	0,45			
Item	Component	Down	Up	Vendor	Serial No.	Gauge OD	Body OD	Body ID	Fish Neck	Comp L (m)	
1	Corehead BHC409Z		HT-30	HC	5287192	8 1/2"	7 1/4"	4"	0,21	0,45	0,45
2	Stabilizer	HT-30	HT-30	BHI	14378317	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	1,67
3	Outertube	HT-30	HT-30	BHI	14348681		7 1/4"	5 15/16"	Slick	7,93	9,60
4	Stabilizer	HT-30	HT-30	BHI	14378315	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	10,82
5	Outertube	HT-30	HT-30	BHI	14278066		7 1/4"	5 15/16"	Slick	7,93	18,75
6	Stabilizer	HT-30	HT-30	BHI	13970760	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	19,97
7	Outertube	HT-30	HT-30	BHI	13956501		7 1/4"	5 15/16"	Slick	7,93	27,90
8	Stabilizer	HT-30	HT-30	BHI	13982992	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	29,12
9	Outertube	HT-30	HT-30	BHI	14477119		7 1/4"	5 15/16"	Slick	7,93	37,05
10	Stabilizer	HT-30	HT-30	BHI	14339941	8 15/32"	7 1/4"	5 15/16"	0,51	1,22	38,27
11	LDA ext.	HT-30	HT-30	BHI	14447792		7 1/4"	5 15/16"	Slick	1,22	39,49
12	Top sub	HT-30	NC50	BHI	14426032		7 1/4"	3 5/32"	Slick	0,76	40,25
13	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39,42	79,67
14	4 x 6 1/2 collar	NC50	NC50	RIG	TALLY					39,39	119,06
15	Jar	NC50	NC50	SLB	TALLY					12,10	131,16
16	3 x 6 1/2 collar	NC50	NC50	RIG	TALLY					27,99	159,15
17	XO	NC50	5 1/2FH	RIG	TALLY					1,00	160,15
18	12 X 5 7/8 HWDP	5 1/2FH	5 1/2FH	RIG	TALLY					110,57	270,72
19	5 7/8 DPS RANGE 3	5 1/2FH	5 1/2FH	RIG	TALLY					13,93	284,65
20											284,65
21											284,65
22											284,65
23											284,65
24											284,65
25											284,65

3. Core Barrel Diagrams

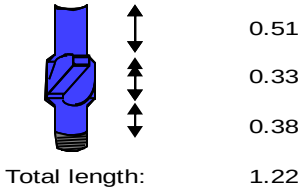
Core Barrel Assembly			
Company: Contractor: Rig Name: Well Name: Company Rep.: Service Eng.:	Equinor Seadrill West Hercules 31/5-7 EOS Jon D. Eikenes K. Larsen / M.Finnesand		
B H I Core Barrel	CORE#	1	
Style: Size: Length: Max Core Length: Weight:	HT-30 7 1/4" x 5 15/16" inch 31.11 meter 27.00 meter 3 ton		
Operating Parameters			
M/U Torque: Yield Torque: Max. Pull: Max. WOB Rotary: Flow Rate: Max dog leg Severity:	30,000 ft/lbs 55,300 ft/lbs 571,000 lbs 12 ton 45-150 rpm 1,000 lpm		
Comments:			
Bit Style: BHC 409 Z Bit Mfr: HC TFA: 1.50 Size: 8 1/2" SN: 7160215 Blades: 9 blades, 9 flow ports. Cutters: 52 x 13mm			

Description	BHA	Length
Top sub		0.77
OT Ext		1.22
Stabilizer		1.22
Outertube		7.93
Stabilizer		1.22
Outertube		7.93
Stabilizer		1.22
Outertube		7.93
Stabilizer		1.22
PDC bit		0.45
Total		31.11

Corehead Specifications:



Stabilizer spec.:



Core Barrel Assembly

Company: **Equinor**
 Contractor: **Seadrill**
 Rig Name: **West Hercules**
 Well Name: **31/5-7 EOS**
 Company Rep.: **Jon D. Eikenes**
 Service Eng.: **K. Larsen /M.Finnesand**

B H I Core Barrel **CORE#** **2**

Style: **HT-30**
 Size: **7 1/4" x 5 15/16"** **inch**
 Length: **58.55** **meter**
 Max Core Length: **54.00** **meter**
 Weight: **6** **ton**

Operating Parameters

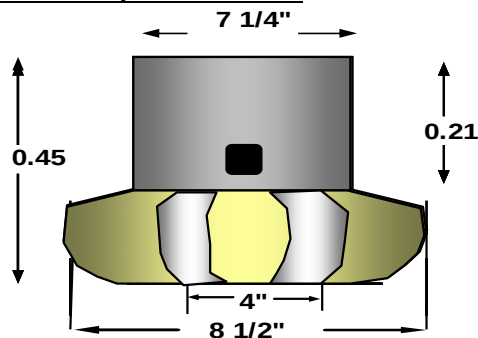
M/U Torque: **30,000** **ft/lbs**
 Yield Torque: **55,300** **ft/lbs**
 Max. Pull: **571,000** **lbs**
 Max. WOB: **12** **ton**
 Rotary: **45-150** **rpm**
 Flow Rate: **1,000** **lpm**
 Max dog leg
 Severity:

Comments:

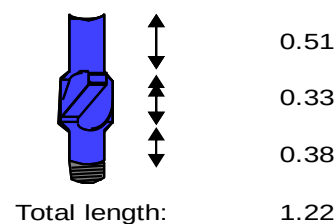
Bit Style: **BHC 409 Z**
 Bit Mfr: **HC**
 TFA: **1.50**
 Size: **8 1/2"**
 SN: **7160215**
 Blades: **9 blades, 9 flow ports.**
 Cutters: **52 x 13mm**

Description	BHA	Length
Top sub		0.76
OT Ext		1.22
Stabilizer		1.22
Outertube		7.93
4 x Stabilizer		4.88
4 x Outertube		31.72
Stabilizer		1.22
Outertube		7.93
Stabilizer		1.22
PDC bit		0.45
Total		58.55

Corehead Specifications:



Stabilizer spec.:



Core Barrel Assembly

Company: Equinor
Contractor: Seadrill
Rig Name: West Hercules
Well Name: 31/5-7 EOS
Company Rep.: Jon D. Eikenes
Service Eng.: K. Larsen /M.Finnesand

B H I Core Barrel **CORE#** 3

Style: HT-30
Size: 7 1/4" x 5 15/16" inch
Length: 86,00 meter
Max Core Length: 81,00 meter
Weight: 9 ton

Operating Parameters

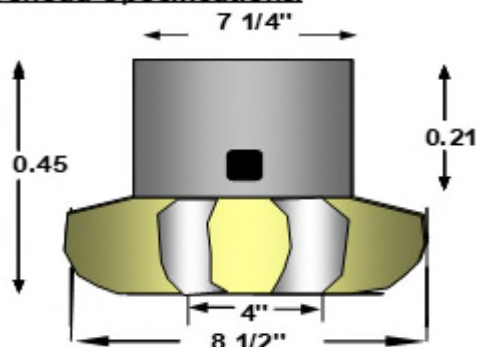
M/U Torque: 30 000 ft/lbs
Yield Torque: 55 300 ft/lbs
Max. Pull: 571 000 lbs
Max. WOB 12 ton
Rotary: 45-150 rpm
Flow Rate: 1 000 lpm
Max dog leg
Severity:

Comments:

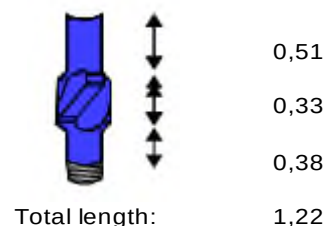
Bit Style: BHC 409 Z
Bit Mfr: HC
TFA: 1,50
Size: 8 1/2"
SN: 5287192
Blades: 9 blades, 9 flow ports.
Cutters: 52 x 13mm

Description	BHA	Length
Top sub		0,76
OT Ext		1,22
Stabilizer		1,22
Outertube		7,93
7 x Stabilizer		8,54
7 x Outertube		55,51
Stabilizer		1,22
Outertube		7,93
Stabilizer		1,22
PDC bit		0,45
Total		86

Corehead Specifications:



Stabilizer spec.:



Core Barrel Assembly

Company: **Equinor**
 Contractor: **Seadrill**
 Rig Name: **West Hercules**
 Well Name: **31/5-7 EOS**
 Company Rep.: **Jon D. Eikenes**
 Service Eng.: **K. Larsen /M.Finnesand**

B H I Core Barrel **CORE#** **4**

Style: **HT-30**
 Size: **7 1/4" x 5 15/16"** **inch**
 Length: **40,25** **meter**
 Max Core Length: **36,00** **meter**
 Weight: **4** **ton**

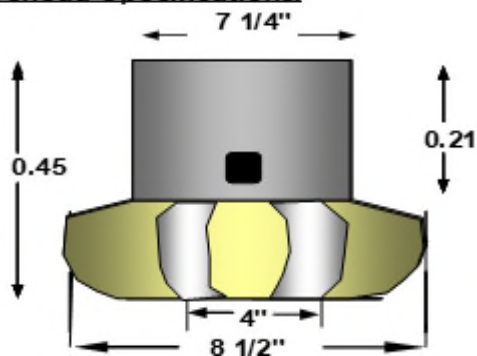
Operating Parameters
 M/U Torque: **30 000** **ft/lbs**
 Yield Torque: **55 300** **ft/lbs**
 Max. Pull: **571 000** **lbs**
 Max. WOB: **12** **ton**
 Rotary: **45-150** **rpm**
 Flow Rate: **1 000** **lpm**
 Max dog leg
 Severity:

Comments:

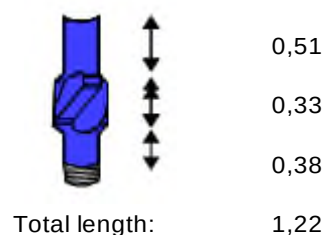
Bit Style: **BHC 409 Z**
 Bit Mfr: **HC**
 TFA: **1,50**
 Size: **8 1/2"**
 SN: **5287192**
 Blades: **9 blades, 9 flow ports.**
 Cutters: **52 x 13mm**

Description	BHA	Length
Top sub		0,76
OT Ext		1,22
Stabilizer		1,22
Outertube		7,93
2 x Stabilizer		2,44
2 x Outertube		15,86
Stabilizer		1,22
Outertube		7,93
Stabilizer		1,22
PDC bit		0,45
Total		40,25

Corehead Specifications:



Stabilizer spec.:

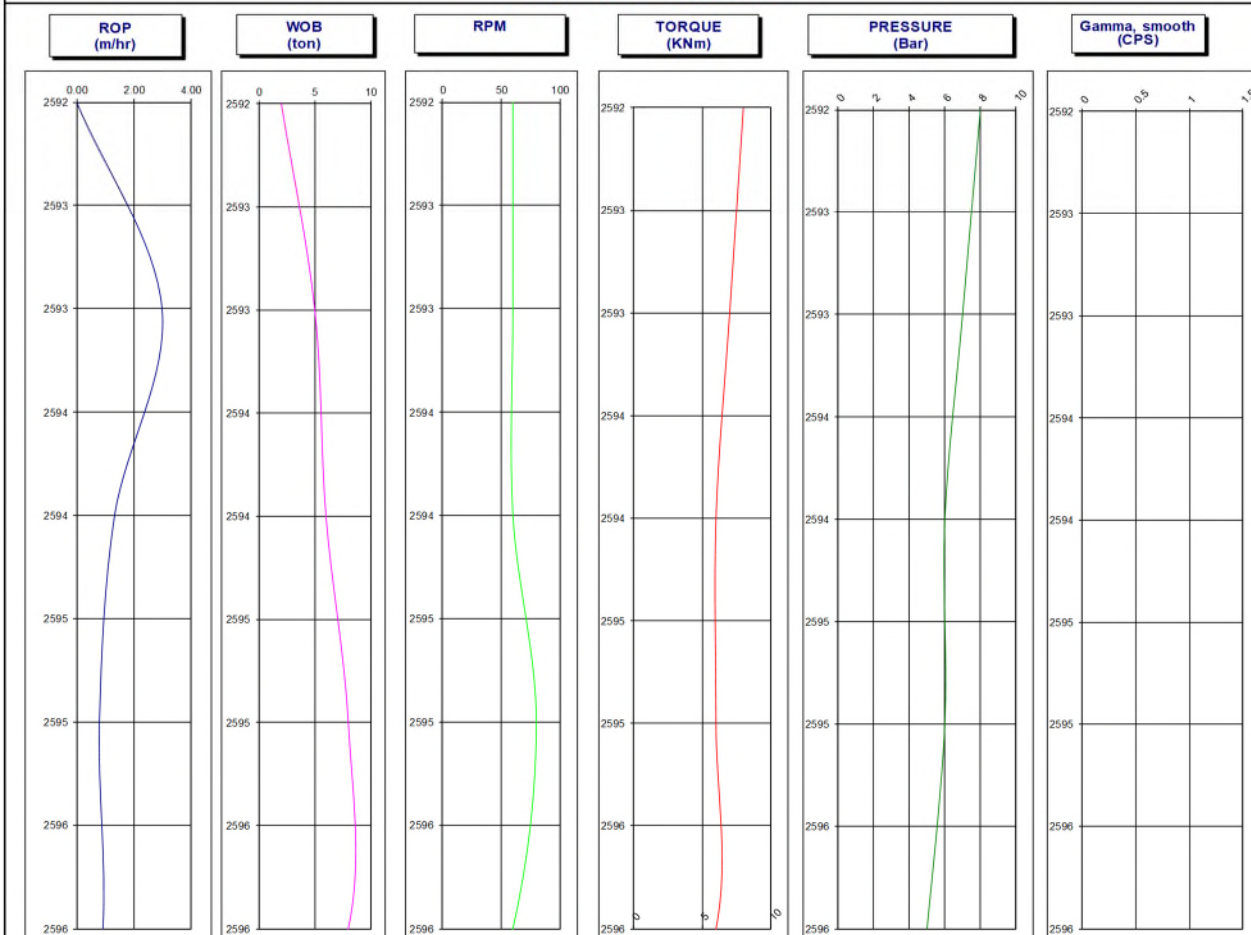


4. Core Logs

CORE LOG

WELL INFORMATION			EQUIPMENT			PERFORMANCE		
Oil Company	Equinor		Core BBL Type	HT-30		Core #	1	
Contractor	Seadrill		Core BBL Size (inches)	7 1/4" x 5 15/16"		Depth In (MD/TVD)	2,592.0 /	
Well Name	31/5-7 EOS		Core BBL Length (mtr)	27.0		Depth Out (MD/TVD)	2,596.0 /	
Field	Northern Lights		Inner Tube Type	Aluminium vented jambuster		Amount Cored (mtr)	4.0	
Rig Name	West Hercules		Catcher Type	Standard spring catcher.		Core Recovery (mtr)	3.2	
Area	North sea					% Recovery	80.0%	
Country	Norway		Stab. Size In (inches)	8 15/32"	Stab. Size Out 8 15/32"	Coring Time (hours)	4.80	
Hole Size	8 1/2"		Core Bit Style	BHC409Z	Core Bit Ser.# 7160215	ROP (mph)	0.8	
Hole Angle	0		Core Bit Gauge OC (inches)	8 1/2"	Core Bit Gauge ID 4"	% Coring Efficiency (Core Cut/ Cbl. Length)	14.8%	
Max DLS (d/30m)	0		C/Bit Shank Type	HT-30	Core Bit TFA 1.50			
Formation	Drake		IADC Grade - In	NEW		Coring Engineer	Kjetil Larsen	
Lithology	Shale		IADC Grade - Out	0-0-BU-A-X-1-PN-PR		Coring Engineer	Magnor Finnesand	
							Massimo	
							31-Dec-19	
Mud Type	WBM	Chlorides 70000	Temperature [deg C]	40	HTHP no	Date		
Mud Weight (S.G.)	1.15	PV 13	LPM	1000	SPM 64	Job No.		
% Oil	N/A	SAND N/A	BAR on bottom	27	Bar off bottom 26	Company Man	Jon Durbeck Elkenes	
% Solids	8.00%	YP 11.5	Casing OD	9 5/8	Casing ID 8.642	Geologist	Mark Vanhathalo	
pH	8	LCM NO	Casing Shoe @ (mtr)	2583.0	Open Hole mtr. 9.0	Geologist	Laure Dieudonne	

OPERATING PARAMETERS



REMARKS :

CORE LOG

WELL INFORMATION

Oil Company	Equinor
Contractor	Seadrill
Well Name	31/5-7 EOS
Field	Northern Lights
Rig Name	West Hercules
Area	North sea
Country	Norway
Hole Size	8 1/2"
Hole Angle	0
Max DLS (d/30m)	0
Formation	Cook
Lithology	Sand/claystone

Mud Type	WBM	Chlorides	67000
Mud Weight (S.G.)	1.15	PV	9
% Oil	NO	SAND	0.1
% Solids	9.00%	YP	8.1
pH	9	LCM	NO

EQUIPMENT

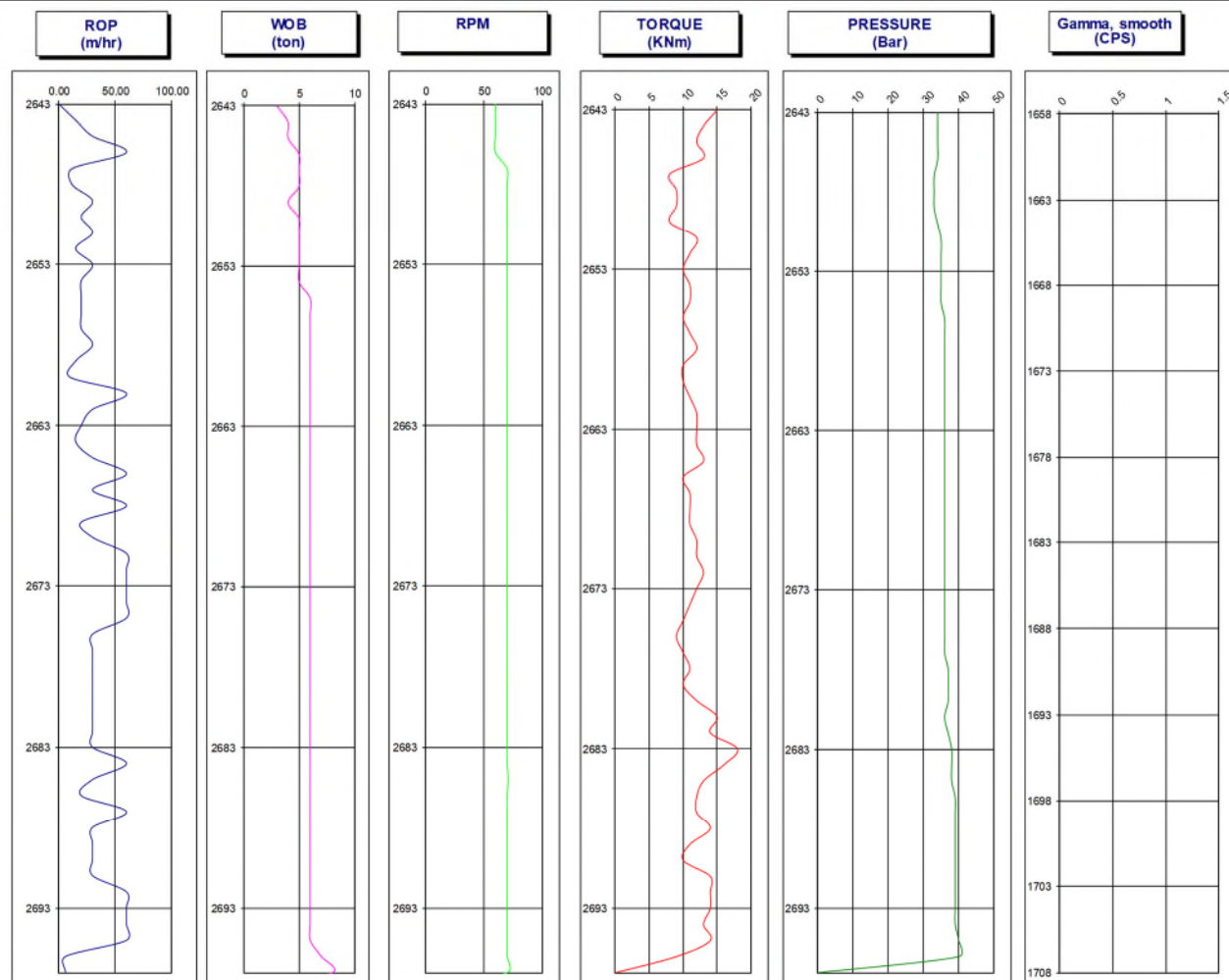
Core BBL Type	HT-30
Core BBL Size (inches)	7 1/4" x 5 15/16"
Core BBL Length (mtr)	54.0
Inner Tube Type	Aluminium vented jambuster
Catcher Type	Standard spring catcher.
Stab. Size In (inches)	8 15/32"
Core Bit Style	BHC409Z
Core Bit Gauge OC (inches)	8 1/2"
C/Bit Shank Type	HT-30
IADC Grade - In	0-0-NO-A-X-I-NO
IADC Grade - Out	1-1-LT-N-X-I-PN-BHA

Temperature [deg C]	8	HTHP	NO
LPM	1000	SPM	52
BAR on bottom	34	Bar off bottom	30
Casing OD	9 5/8"	Casing ID	8.642
Liner Shoe @ (mtr)	2583.0	Open Hole mtr.	60.0

PERFORMANCE

Core #	2
Depth In (MD/TVD)	2,643.0 /
Depth Out (MD/TVD)	2,697.0 /
Amount Cored (mtr)	54.0
Core Recovery (mtr)	53.4
% Recovery	98.9%
Coring Time (hours)	2.45
ROP (mph)	22.0
% Coring Efficiency (Core Cut/ Cbl. Length)	100.0%
Coring Engineer	Kjetil Larsen
Coring Engineer	Magnor Finnesand
Coring Engineer	Massimo
Date	01.01.2020
Job No.	110140287
Company Man	Roy Nordstokke
Geologist	Mark Vanhatalo
Geologist	Richard Day

OPERATING PARAMETERS

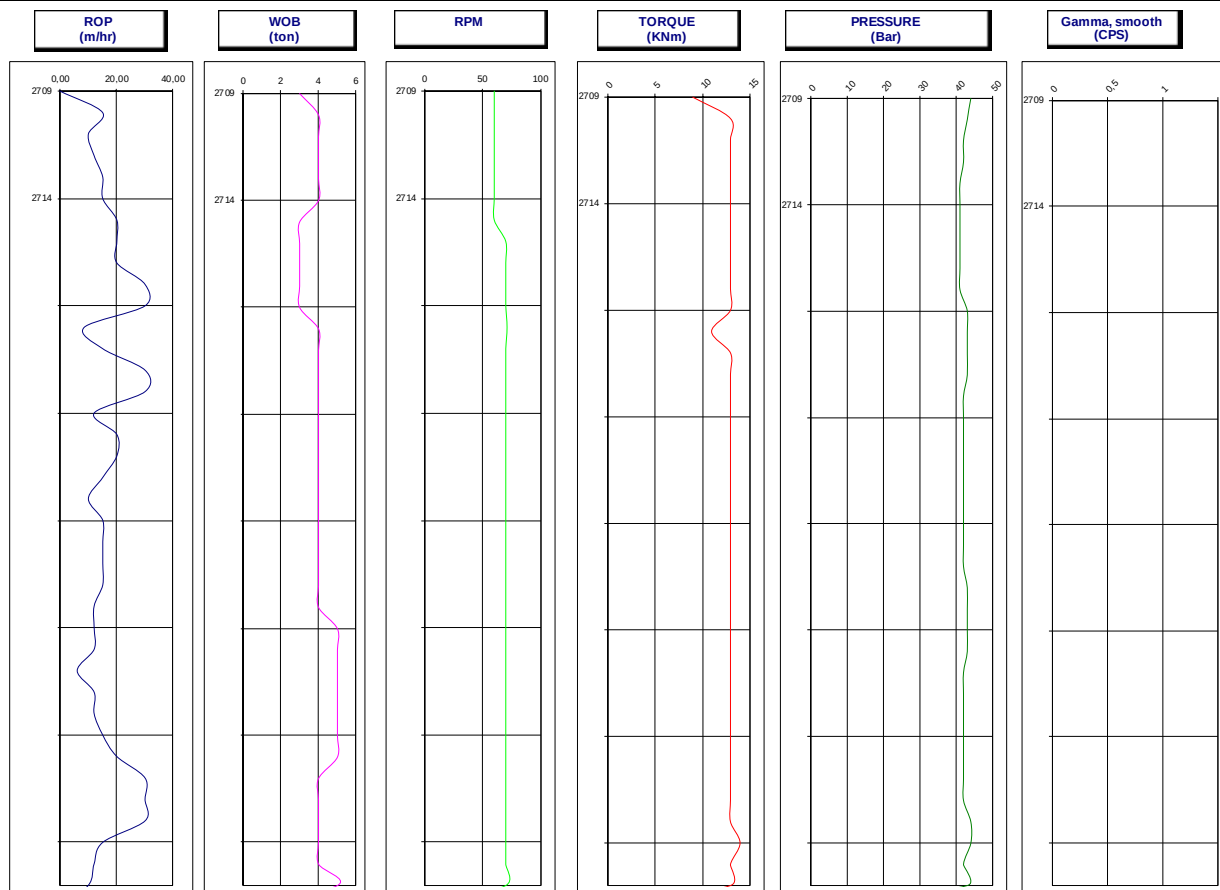


REMARKS :

CORE LOG

WELL INFORMATION			EQUIPMENT			PERFORMANCE			
Oil Company	Equinor		Core BBL Type	HT-30		Core #	3		
Contractor	Seadrill		Core BBL Size (inches)	7 1/4" x 5 15/16"		Depth In (MD/TVD)	2 709,0 /		
Well Name	31/5-7 EOS		Core BBL Length (mtr)	81,0		Depth Out (MD/TVD)	2 746,0 /		
Field	Northern Lights		Inner Tube Type	Aluminium vented jambuster		Amount Cored (mtr)	37,0		
Rig Name	West Hercules		Catcher Type	Standard spring catcher.		Core Recovery (mtr)	35,3		
Area	North sea					% Recovery	95,4%		
Country	Norway		Stab. Size In (inches)	8 15/32"	Stab. Size Out	8 15/32"	Coring Time (hours)	2,50	
Hole Size	8 1/2"		Core Bit Style	BHC409Z	Core Bit Ser.#	5287192	ROP (mph)	14,8	
Hole Angle	0		Core Bit Gauge OD (inches)	8 1/2"	Core Bit Gauge ID	4"	% Coring Efficiency (Core Cut/ Cbl. Length)	45,7%	
Max DLS (d/30m)	0		C/Bit Shank Type	HT-30	Core Bit TFA	1,50			
Formation	Johansen		IADC Grade - In	NEW		Coring Engineer	Kjetil Larsen		
Lithology	Sand/claystone		IADC Grade - Out	0-0-WT-A-X-I-PN-PR		Coring Engineer	Magnor Finnesand		
							Massimo		
Mud Type	WBM	Chlorides	65800	Temperature [deg C]	5	HTHP	NO	Date	10.01.2020
Mud Weight (S.G.)	1,15	PV	8	LPM	1000	SPM	54	Job No.	110140287
% Oil Glycol	N/A	SAND	0,2	BAR on bottom	44	Bar off bottom	39	Company Man	Roy N
% Solids	9,00 %	YP	10,1	Casing OD	9 5/8"	Casing ID	8,681	Geologist	Anders Grev Hesthagen
pH	9	LCM	NO	Liner Shoe @ (mtr)	2583,0	Open Hole mtr.	127,0	Geologist	Maqsood Ahmed

OPERATING PARAMETERS

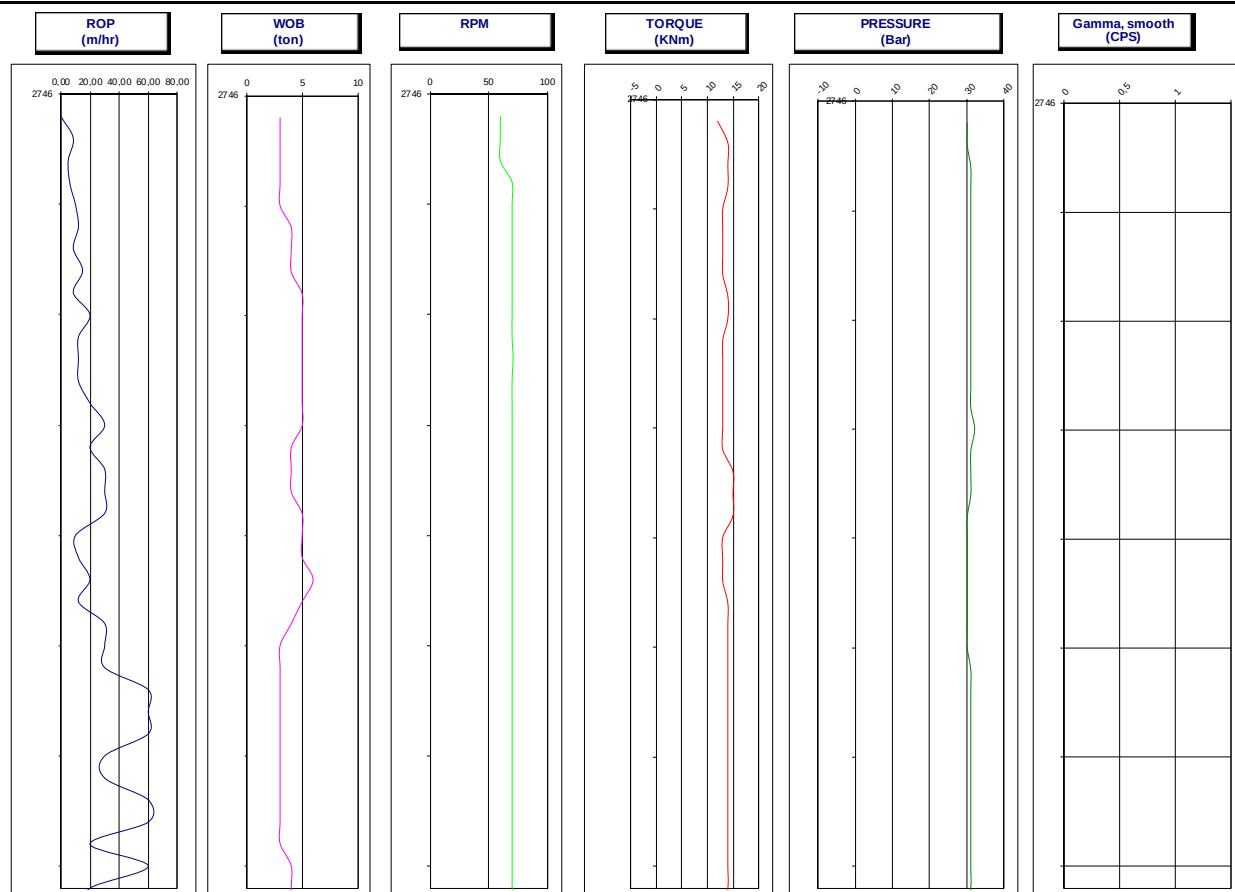


REMARKS : _____

CORE LOG

WELL INFORMATION				EQUIPMENT				PERFORMANCE					
Oil Company	Equinor			Core BBL Type	HT-30			Core #	4				
Contractor	Seadrill			Core BBL Size	(inches)	7 1/4" x 5 15/16"			Depth In	(MD/TVD)	2 747,0 /		
Well Name	31/5-7 EOS			Core BBL Length	(mtr)	36,0			Depth Out	(MD/TVD)	2 783,7 /		
Field	Northern Lights			Inner Tube Type	Aluminium vented jambuster			Amount Cored	(mtr)	36,7			
Rig Name	West Hercules			Catcher Type	Standard spring catcher.			Core Recovery	(mtr)	33,9			
Area	North sea			Stab. Size In	(inches)	8 15/32"	Stab. Size Out	8 15/32"	% Recovery	92,4%			
Country	Norway					BHC409Z	Core Bit Ser.#	5287192	Coring Time	(hours)	2,25		
Hole Size	8 1/2"			Core Bit Style	(inches)	8 1/2"	Core Bit Gauge ID	4"	ROP	(mph)	16,3		
Hole Angle	1			Core Bit Gauge OD		HT-30	Core Bit TFA	1,50	% Coring Efficiency	101,9%			
Max DLS (d/30m)	0			C/Bit Shank Type	0-0-WT-A-X-I-NO			(Core Cut/ Cbl. Length)					
Formation	Johansen			IADC Grade - In	0-0-LT-N-X-I-PN-PR			Coring Engineer	Kjetil Larsen				
Lithology	Sand/claystone			IADC Grade - Out				Coring Engineer	Magnor Finnesand				
Mud Type	WBM	Chlorides	65800	Temperature [deg C]	5	HTHP	NO	Date	13.01.2020				
Mud Weight (S.G.)	1,15	PV	8	LPM	1000	SPM	54	Job No.	110140287				
% Oil Glycol	N/A	SAND	0,2	BAR on bottom	31	Bar off bottom	30	Company Man	Roy Nordstoke				
% Solids	9,00 %	YP	10,1	Casing OD	9 5/8"	Casing ID	8,681	Geologist	Anders Grev Hesthagen				
pH	9	LCM	NO	Liner Shoe @	(mtr)	2583,0	Open Hole mtr.	263,0	Geologist	Magsood Ahmed			

OPERATING PARAMETERS



REMARKS :

5. TECHNICAL REPORT

Tool box talks were held with the drilling / deck crews prior to all core barrel handling operations.

CORE RUN # 1:

2592m MD to 2596m MD

Formation: Drake

Lithology: Shale

30 December 2019

A pre-torqued, 27m x 7 1/4" x 5 15/16" HT-30 Max Jam Buster core barrel was picked up from the pipe deck. The assembly was dressed with pressure vented aluminum inner tubes and a 8 1/2" x 4" Quick Core BHC 409Z pdc (# 7160215) core head. The inner barrels were spaced out to 2.1cm to ensure that there was sufficient clearance for thermal expansion due to down hole temperature. This allowed the outer barrel to rotate independently of the inner barrel arrangement. Down hole temperature was approximately 50°C.

31 December 2019

RIH to one stand off-bottom, washed down with 1500 LPM. Tagged bottom and circulated 2 x string volume with 1500 LPM. Dropped the ball and pumped down with 1000 LPM and 15 bar pressure. The pressure increased by 5 bar once the ball seated. Took TQ and SPP parameter readings off bottom.

Coring commenced at 2592m MD with surface parameters of 1000 LPM, 60 RPM and 3-12 ton WOB. Cored 4m slow start from the beginning and a low penetrating rate. The pressure gradually increased by 5 bar and torque was lost. Dropped down to 2-3 kN•m and no ROP. After discussions with the geologist it was decided to pull off.

Pulled off bottom with 9 ton over pull observed.

Pulled wet into the shoe approximately 10 meters.

Circulated bottoms up with 1200 LPM and boosted the riser.

Flow checked and then started to POOH according to Equinor drilling program.

Bit time was 5 hours and ROP 0.8 m/hr. A total of 4m was cored.

31 December 2019

At surface a toolbox talk was conducted with the drilling and deck crew where safety was highlighted as the overriding requirement for all concerned. The crews were instructed on how to lay out the core safely. Emphasis was made regarding the necessity for careful handling of the core barrels.

The two empty inner tubes were laid out on the pipe deck and the bottom inner barrel with core inside was laid over to the cutting area.

Retrieved the 1 1/4" ball and float from the core assembly and laid out the core according to Equinor and Baker Hughes procedures.

The BHC409Z bit (#7160215) was inspected. 7 ports were plugged and balled up.

The corehead graded 0-0-BU-A-X-I-PN-PR.

The Jam Buster was not activated.

Cored 4 meter and recovered 3.2m, 80% recovery.

Core run#1 was filed and preserved with baseoil.

CORE RUN # 2

2643m MD to 2697m MD
Formation: Cook
Lithology: Sand/Claystone

1 January 2020

A pre-torqued, 54m x 7 1/4" x 5 15/16" HT-30 Max Jam Buster core barrel was picked up, from the pipe deck. The assembly was dressed with pressure vented aluminum inner tubes and the 8 1/2" x 4" Quick Core BHC 409Z pdc (# 7160215) core head was re-run. The inner barrels were spaced out to 3.1 cm to ensure that there was sufficient clearance for thermal expansion due to down hole temperature. This allowed the outer barrel to rotate independently of the inner barrel arrangement. Down hole temperature was approximately 50°C.

RIH to one stand off-bottom, washed down with 1500 LPM. Tagged bottom and circulated 2 x string volume with 1500 LPM. Dropped the ball and pumped down with 1000 LPM and 20 bar pressure. The pressure increased by 10 bar once the ball seated. Took TQ and SPP parameter readings off bottom.

Coring commenced at 2643m MD with surface parameters of 1000 LPM, 60-70 RPM and 3-6 ton WOB. Cored 54m with one connection and steady parameters.

Pulled off bottom with 14 ton over pull observed.

Pulled wet into the shoe.

Circulated bottom up with 1200 LPM.

Flow checked and then started to POOH according to Equinor drilling program.

Bit time was 3 hours and ROP 22 m/hr. A total of 54m was cored.

2. January 2020

At surface a toolbox talk was conducted with the drilling and deck crew where safety was highlighted as the overriding requirement for all concerned. Due to bad weather with a lot of rig movement this was especially highlighted. The crews were instructed on how to lay out the core safely. Emphasis was made regarding the necessity for careful handling of the core barrels.

The two empty inner tubes were laid out on the pipe deck and the bottom inner barrel with core inside was laid over to the cutting area. Retrieved the 1 1/4" ball and float from the core assembly and laid out the core according to Equinor and Baker Hughes procedures.

The BHC409Z bit (#7160215) was inspected. 3 ports were plugged and several teeth were chipped or lost.

The corehead was graded 1-1-LT-N-X-I-PN-BH.

Jam Buster was not activated.

Cored 54m and recovered 53.4m, 98.9% recovery.

CORE RUN # 3

2709m MD to 2746m MD
Formation: Johansen
Lithology: Sand/Claystone

10-11 January 2020

A pre-torqued, 81m x 7 1/4" x 5 15/16" HT-30 Max Jam Buster core barrel was picked up, from the pipe deck and derrick. The assembly was dressed with pressure vented aluminum inner tubes and a 8 1/2" x 4" redressed new core bit BHC 409Z pdc (# 5287192). The inner barrels were spaced out to 5.1 cm to ensure that there was sufficient clearance for thermal expansion due to down hole temperature. This allowed the outer barrel to rotate independently of the inner barrel arrangement. Down hole temperature was approximately 55°C.

RIH to one stand off-bottom, washed down with 1500 LPM. Tagged high 4m at 2705m MD and washed down careful with same rate and 5–10 RPM. Observed a 10 bar pressure increase but no additional weigh. Circulated 2 x string volume with 1500 LPM to allow the inner barrel to cool down. Afterwards the bottom was tagged at 2709 torque and 5 ton WOB to confirm. Dropped the ball and pumped down with 1000 LPM and 20 bar pressure. The pressure increased by 18 bar once the ball seated. Took TQ and SPP parameter readings off bottom.

Coring commenced at 2709m MD with surface parameters of 1000 LPM, 60-70 RPM and 3-6 ton WOB. Cored for 37m.

The heave prior to connection was 2.8m. After making the connection the parameters changed; torque dropped to off-bottom and the pressure dropped by 10 bar and did it was not possible to core any further. The geologist was on the rig floor and following a conversation with the company man it was decided to POOH. No over pull was observed and it was confirmed that the Jam Buster system had been activated.

Pumped into the shoe with 1200 LPM.

Circulated bottom up with 1200 LPM.

Flow checked and then started to POOH according to Equinor drilling program.

Bit time was 2.5 hours and ROP 14.8 m/hr. A total of 37m was cored.

12. January 2020

At surface a toolbox talk was conducted with the drilling and deck crew where safety was highlighted as the overriding requirement for all concerned. Due to bad weather with a lot of rig movement this was especially highlighted. The crews were instructed on how to lay out the core safely. Emphasis was made regarding the necessity for careful handling of the core barrels.

There was a lot of rig moment so laydown frames were used for the inner barrel with the core inside laid over to cutting area. Retrieved the 1 1/4" ball and float from the core assembly and laid out the core according to Equinor and Baker Hughes procedures.

The BHC409Z bit (# 5287192) was inspected. 5 ports were plugged.

The corehead was graded 0-0-WT-A-X-I-PN-P.

One Jam buster was activated at approximately 2735m MD

Cored 37m and recovered 35.3m, 95.4% recovery.

CORE RUN # 4

2747m MD to 2784m MD
Formation: Johansen
Lithology: Sand/Claystone

12-13 January 2020

Remodelled the 81m core barrel from run #3 to a 36m x 7 1/4" x 5 15/16" HT-30 Max Jam Buster. The assembly was dressed with pressure vented aluminum inner tubes and the 8 1/2" x 4" Quick Core BHC 409Z pdc (# 5287192) core head was re-run. The inner barrels were spaced out to 2.9 cm to ensure that there was sufficient clearance for thermal expansion due to down hole temperature. This allowed the outer barrel to rotate independently of the inner barrel arrangement. Down hole temperature was approx. 60°C.

RIH to one stand off-bottom, washed down with 1500 LPM. Washed down and tagged the bottom at 2746m. Circulated 2 x string volume with 1500 LPM to flush through the inner barrel. Torque from 3-10 tq and 5 ton WOB to confirm. Dropped the ball and pumped down with 1000 LPM and 19 bar pressure. The pressure increased by 28 bar once the ball seated. Took TQ and SPP parameter readings off bottom.

Coring commenced at 2746m MD with surface parameters of 1000 LPM, 60-70rpm and 3-6ton WOB. Cored For 36m. Max heave was 2.8m. No over pull was observed when POOH.

Bit time was 2.5 hours and ROP 16.3 m/hr. A total of 36.7m was cored.

At surface a toolbox talk was conducted with the drilling and deck crew where safety was highlighted as the overriding requirement for all concerned. Due to bad weather with a lot of rig movement this was especially highlighted. The crews were instructed on how to lay out the core safely. Emphasis was made regarding the careful handling of the core barrels.

There was a lot of rig moment so laydown frames were used for the inner barrel with the core inside laid over to cutting area. Retrieved the 1 1/4" ball and float from the core assembly and laid out the core according to Equinor and Baker Hughes procedures.

The BHC409Z bit (# 5287192) was inspected. 3 ports were plugged.

The core head was graded 0-0-LT-A-X-I-PN-BH.

One Jam Buster was activated at approximately 2750m MD.

Cored 36.7m and recovered 33.9m, 92.4% recovery.

6. Performance Review

Safety

- The BHI coring specialist conducted “Tool Box Talks” to familiarise drill and deck crews with the equipment and processes involved.
- No safety related incidents occurred during the coring operation.

Core Performance

- It is believed that the cores are of high quality and will enable a full suite of analysis and geological interpretation to satisfy the requirements of a typical analysis program.
- No lay down, flexing induced damage was apparent and the cores laydown frames supplied were used for all core movements.
- The Jam Buster system was activated two times; on run #3 it was activated at approximately 2735m MD and on run #4 at approximately 2750m MD.
- The use of pre-torqued barrels reduced handling and saved rig time during make-up of core barrel